Lesson Planning

Complete Lesson Plan Book

Class: Class 11

Subject: Chemistry

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Atomic Structure: Electrons and Energy Levels

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Unit 1: Atomic Structure: Electrons and Energy Levels

Learning Objectives

• Students will describe how electrons are arranged in energy levels around the nucleus.

• Students will write the electronic configuration of the first twenty elements.

• Students will explain why energy levels fill in a fixed order using the 2, 8, 8 pattern.

Key Concepts

• An atom has a tiny nucleus of protons and neutrons, with electrons moving in energy levels, or shells, around it.

• The first shell holds at most 2 electrons, the second 8, and the third 8 for the first twenty elements.

• Electrons fill the lowest available energy level first, which is why configurations follow a regular pattern.

• The number of electrons in the outer shell decides how an element reacts, since atoms gain, lose or share outer electrons.

Teaching Methodology (Step by Step)

1. Ask why sodium reacts violently while neon does not, and link the answer to outer electrons.
2. Draw the Bohr diagram of an atom on the board, labelling nucleus and shells.
3. Work out the configurations of carbon, oxygen and calcium together.
4. Show how the outer-shell count predicts valency with two examples.
5. Students write configurations for five elements and check each other's outer shells.

Classroom Activities

• Shell race: groups build Bohr diagrams for assigned elements from cards.

• Configuration quiz: students match elements to their configurations.

• Predict the reaction: students predict valency from a given configuration.

Assessment Questions

1. How many electrons can the first and second shells hold?
2. Write the electronic configuration of magnesium (atomic number 12).
3. Why do elements in the same group behave similarly?

Homework

• Write configurations for the first twenty elements in a table.

• Draw Bohr diagrams for sodium and chlorine and explain why they combine.

Time Allocation

40 minutes